



Konstruct

Upload a building plan. Get a priced material list and the phone numbers of vendors near your site.

Project documentation for the 300-level presentation

Next.js 16, PostgreSQL, Drizzle, one AI call per plan

Live app: <https://konstruct-production-c62c.up.railway.app>

Demo login: demo@konstruct.ng / demo1234

Contents

1. Overview
2. How the core logic works
3. Architecture and data model
4. Walkthrough of every screen
5. Mobile view
6. The PDF breakdown
7. Sample plans and AI results
8. Running locally
9. Testing
10. Deployment
11. 5-minute presentation script
12. Decisions and limits

1. Overview

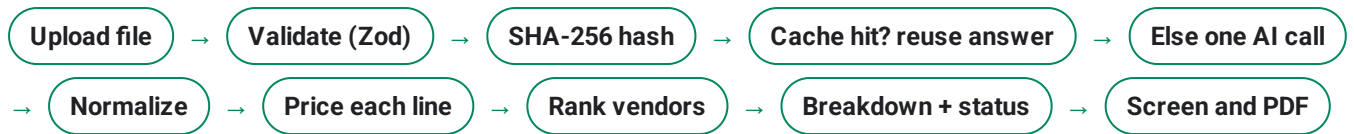
People building in Nigeria usually get a drawing from an architect and then spend days working out how many bags of cement, blocks and rods to buy, and who sells them nearby. Konstruct does that first pass in under a minute.

What it does: a client uploads a building plan (PDF or photo). One AI call reads the drawing and returns a list of materials with quantities. Plain code then prices every line from vendors near the site, ranks the vendors, and produces a professional PDF that the client can share or print. Each line has call and WhatsApp buttons for the vendors who stock it.

Core features

- Upload a plan (PDF, PNG, JPG or WEBP up to 8 MB) or try one of five sample drawing sheets.
- One AI call per file. The answer is cached by the file's SHA-256 hash, so the same file is never sent twice.
- Material breakdown in Naira grouped into 11 categories, with a 10% contingency.
- 34 listed vendors in 8 cities (Lagos, Ikorodu, Ibadan, Abuja, Port Harcourt, Benin City, Enugu, Kano) located in real building-material markets, stocking real brands such as Dangote 3X, BUA, Lafarge Elephant, Nigerchin and Dulux.
- Vendors ranked by distance tier: same city, same state, same geopolitical zone.
- Statuses: Ready, Needs review (a line has no price or was hard to measure), Prices stale (priced over 30 days ago).
- Professional PDF estimate with bill of materials, totals, vendor contacts and assumptions.
- Works on phones: bottom navigation, tap-to-call, no sideways scrolling.

2. How the core logic works



The one AI call (`src/lib/ai.ts`, `src/lib/extraction.ts`)

The file is sent to OpenRouter (model google/gemini-3.8-flash) together with a short prompt and a compact catalog of 46 material codes, one line each (CEM-50|Portland cement, 50kg bag|bag). The request uses a strict JSON schema as the response format, so the reply is always the same shape: title, building type, floors, bedrooms, floor area, summary, assumptions and a list of items with code, quantity, unit, confidence, basis. A typical plan costs about 2,500 input tokens and 3,500 output tokens (about USD 0.015) and takes about 20 seconds with reasoning effort set to low (OPENROUTER_REASONING).

The model is asked only to measure. It does not price anything or choose vendors, so the token count stays small and every number on screen that involves money comes from our own database.

Normalizing the answer

- Unknown codes become OTHER. OTHER lines that clearly name a catalog item (for example "louvre window") are moved onto that item by a short alias list.
- Catalog names and units always win over the model's wording. Countable units (bags, pieces, lengths) are rounded up; bulk units (tonnes, m2) keep one decimal.
- Duplicate codes are merged and keep the weakest confidence. Zero or negative lines are dropped. Em dashes, curly quotes and emoji are stripped.
- If the model says the file is not a building plan, the upload is rejected with an inline message and nothing is cached.

Pricing and vendor ranking (`src/lib/vendors.ts`, `src/lib/breakdown.ts`)

Rule	Detail
Distance tier	0 same city, 1 same state, 2 same geopolitical zone, 3 elsewhere. Tier 3 vendors are never suggested.
Vendor order	Nearest tier first, then cheapest, then highest rating.
Line price	Cheapest vendor in the nearest tier that stocks the item. If no vendor nearby: the catalog reference price. OTHER items: no price until the user types one.
Totals	Line amount = quantity x unit price. Subtotal + 10% contingency = estimated total. Unpriced lines are listed but excluded.
Price freshness	Current up to 14 days (the quote validity), ageing to 30 days, stale after 30 days.
Status	Needs review if any line is unpriced or low confidence; else Prices stale if stale; else Ready.
Quote reference	KON-YYYYMMDD-NNNN from the pricing date and plan id.

Prices are stored on each line when the plan is priced, so an old estimate keeps its old numbers. Refresh prices re-runs only the pricing step with today's vendor price lists; it never calls the AI again.

Every business rule is a pure function in `src/lib/` that takes "today" as an argument. Setting `KONSTRUCT_TODAY=2026-10-04` pins the date, which is how tests, screenshots and the demo stay the same every time.

3. Architecture and data model

Layer	Choice
Web framework	Next.js 16 App Router, React 19 Server Components and Server Actions, TypeScript strict
Database	PostgreSQL with Drizzle ORM; versioned SQL migrations made by drizzle-kit in drizzle/
Validation	Zod schemas shared by all forms; errors returned per field and shown inline
Auth	Email and password (bcryptjs), HS256 JWT signed with jose in an HTTP-only cookie, 7 days. src/proxy.ts redirects signed-out visitors (Next 16 renamed middleware to proxy)
AI	OpenRouter chat completions with a file part and a strict JSON schema; mock mode for tests
PDF	@react-pdf/renderer with Roboto and Google Sans embedded from assets/fonts/
UI	Plain CSS styled after the Google Play Store, Google Sans and Roboto from @fontsource, lucide-react icons, DiceBear initials avatars generated on the server, custom SVG logo and illustrations
Hosting	Railway: one Node service built with Railpack and one PostgreSQL service

Folders

```
src/app/           pages, route handlers and server actions
  (auth)/signin, signup  (app)/plans, plans/new, plans/[id], vendors, account
  plans/[id]/pdf plans/[id]/file api/health
src/lib/           pure business rules: catalog, extraction, vendors, breakdown, dates,
money, validation
src/db/            schema, queries (every plan query takes userId), seed data
src/server/        session cookies, upload pipeline, PDF document
scripts/           migrate, seed, make-samples, extract-samples, docs/
tests/             unit, integration (real Postgres), e2e (Playwright)
```

Tables

Table	Columns (main)	Notes
users	id, name, email (unique), password_hash, city, state	City sets the default site and vendor list
vendors	id, name, phone, whatsapp, area, address, city, state, rating, reviews, verified, delivers, categories[], brands	Pre-listed suppliers
vendor_prices	vendor_id, code, unit_price, updated_on	One price per vendor per material
plans	id, user_id, title, building_type, city, state, floors, bedrooms, floor_area_m2, summary, assumptions, file_name, file_hash, ai_model, sample_key, priced_on, created_on	Owned by one user
plan_items	id, plan_id, position, code, description, quantity, unit, confidence, basis, unit_price, vendor_id	Price snapshot per line

plan_files	plan_id, mime, bytes	The uploaded drawing
extraction_cache	file_hash, model, result, created_at	Raw AI answer per file

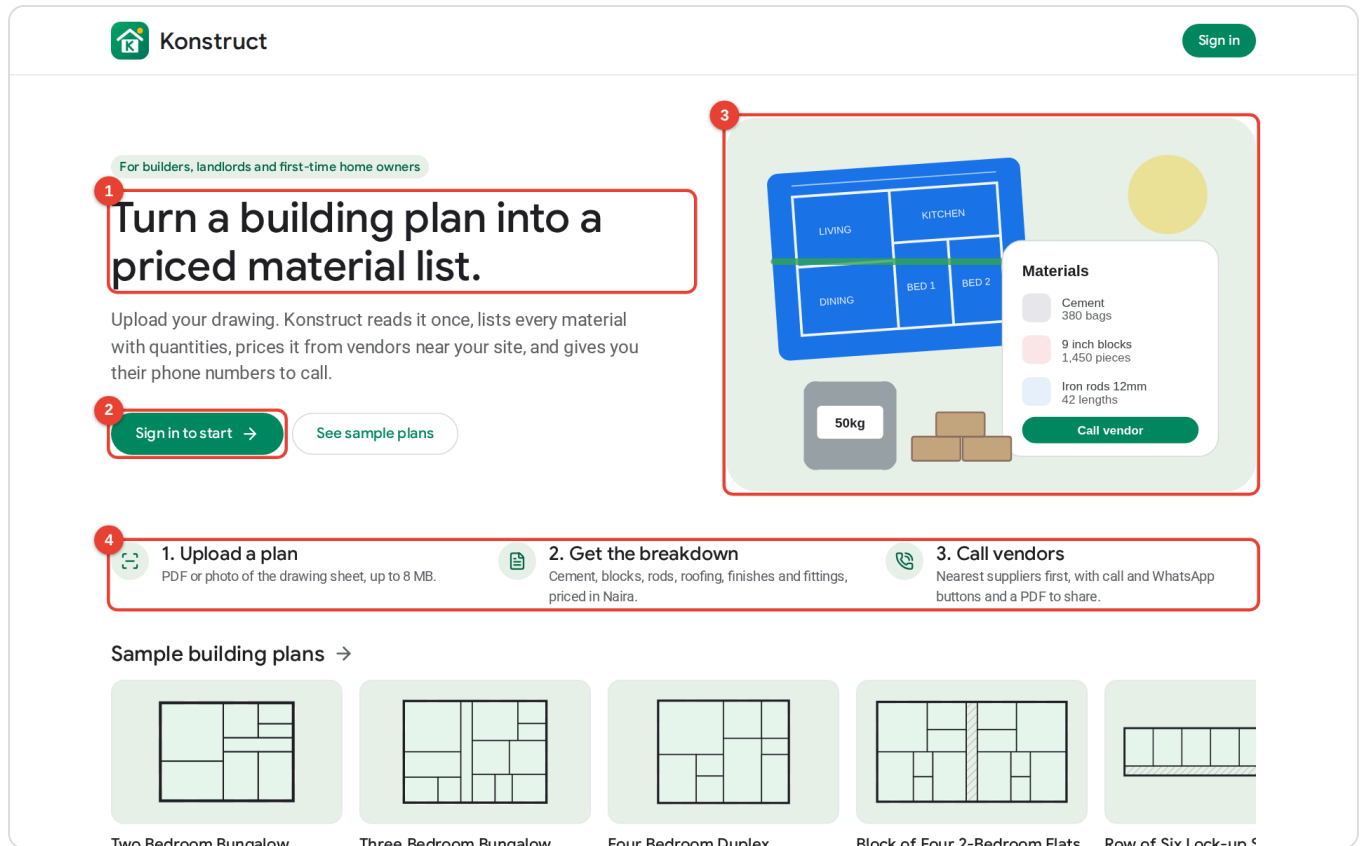
Scoping: every query that reads or changes a plan, its items or its file joins on `plans.user_id = signed-in user`. Integration and e2e tests check that a second user gets "not found" for the first user's plan, PDF and file.

4. Walkthrough of every screen

Screenshots are taken by `scripts/docs/build-docs.ts` with Playwright on a 1280 x 800 window. The red numbers match the notes under each picture.

1. Landing page

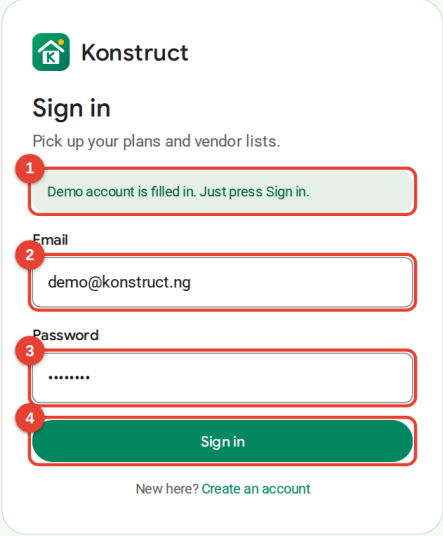
The public page explains the idea in one sentence and shows the five sample plans.



- 1 The promise in plain words: a plan in, a priced material list out.
- 2 Sign in to start. Signed-in users see Open my plans instead.
- 3 Custom SVG illustration stored in public/, no image CDN.
- 4 Three steps: upload, get the breakdown, call vendors.

2. Sign in

The demo account is pre-filled so the presenter only presses one button.



The image shows a 'Sign in' form for 'Konstruct'. The form is centered on a light green background. It has a white background with rounded corners. At the top is the 'Konstruct' logo (a green house icon with a white 'K') and the text 'Konstruct'. Below the logo is the title 'Sign in' and a subtitle 'Pick up your plans and vendor lists.'.

The form contains four main elements, each with a red circular number in the top-left corner:

- 1** A green message box with the text 'Demo account is filled in. Just press Sign in.'
- 2** An 'Email' input field containing the text 'demo@konstruct.ng'.
- 3** A 'Password' input field containing seven dots '.....'.
- 4** A green 'Sign in' button.

Below the 'Sign in' button is a link that says 'New here? Create an account'.

- 1** Tells the viewer the demo login is already filled in.
- 2** demo@konstruct.ng, filled as the default value.
- 3** demo1234, also filled in.
- 4** Signs in and sets a signed JWT in an HTTP-only cookie.

3. Create account with inline errors

Zod validates on the server and each message appears under its own field.

The screenshot shows a 'Create your account' form for 'Konstruct'. The form includes fields for 'Full name', 'Email', 'Password', and 'Your city'. Each field has a red border and an inline error message below it. The 'Full name' field has the error 'Enter your full name'. The 'Password' field has the error 'Use at least 8 characters'. The 'Your city' field has the error 'Choose a city'. A green 'Create account' button is at the bottom, and a link to 'Sign in' is below it. Three red circles with numbers 1, 2, and 3 point to the error messages for the 'Full name', 'Password', and 'Your city' fields respectively.

Konstruct

Create your account

We use your city to suggest vendors near you.

Full name

Enter your full name

Email

Enter your email

Password

Use at least 8 characters

Your city

Choose a city

Choose a city

Create account

Have an account? [Sign in](#)

- 1 Inline error under the field that is wrong.
- 2 Password rule: at least 8 characters.
- 3 City drives which vendors are suggested first.

4. My plans

Every saved plan, like the Play Store list. The status chip says what to do next.

Konstruct My plans Upload Vendors

My plans
Welcome back, Chidinma. 5 plans saved.

1 **All** 5 Ready 3 Needs review 1 Prices stale 1

2

3

4

5

6 + Upload a plan

Nwosu family duplex, L...
Duplex in Lagos
Ready 42 items
Priced yesterday
₦66.7M

Ogui Road shops
Commercial in Enugu
Needs review 37 items
Priced 3 days ago
₦21.2M

Dr Bello bungalow, Lugbe
Bungalow in Abuja
Ready 42 items
Priced 9 days ago
₦32M

Rumuokoro flats
Block of flats in Port Harcourt
Ready 43 items
Priced 22 days ago
₦83.7M

Ogunleye starter home
Bungalow in Ikorodu
Prices stale 41 items
Priced 46 days ago
₦16.6M

Try a sample plan →
Real-style drawing sheets from Lagos, Abuja, Port Harcourt and Enugu. Already read by the AI, so they open instantly.

Two Bedroom Bungalow
Ikorodu, Lagos
Compact family starter home, about 96 m2, single floor.
Try this plan

Three Bedroom Bungalow
Abuja, FCT
Three bedrooms with master en-suite and study, about 162 m2.
Try this plan

Four Bedroom Duplex
Lagos, Lagos
Two-storey family home, suspended slab, about 328 m2.
Try this plan

Block of Four 2-Bedroom Flats
Port Harcourt, Rivers
Ground plus one, two flats per floor, about 440 m2.
Try this plan

Row of Six Lock-up
Enugu, Enugu
Six shops with front walk-toilets, about 144 m2.
Try this plan

- 1 Filter chips: All, Ready, Needs review, Prices stale, with counts.
- 2 A plan: floor plan icon, building type, city and status.
- 3 Needs review: some lines have no price yet.
- 4 Prices stale: priced more than 30 days ago.
- 5 Estimated total including 10% contingency.
- 6 Upload a new plan.

5. Try a sample plan

Five sample drawing sheets. Their AI answers are cached, so they open in about a second.



My plansUploadVendors

Welcome back, Chidinma. 5 plans saved.

CO

All 5Ready 3Needs review 1Prices stale 1



Nwosu family duplex, L...
Duplex in Lagos
Ready 42 items
Priced yesterday
₦66.7M



Ogui Road shops
Commercial in Enugu
Needs review 37 items
Priced 3 days ago
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Dr Bello bungalow, Lugbe
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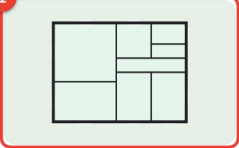


Ogunleye starter home
Bungalow in Ikorodu
Prices stale 41 items
Priced 46 days ago
₦16.6M

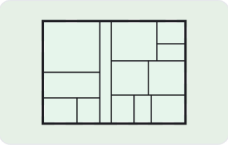
Try a sample plan →

Real-style drawing sheets from Lagos, Abuja, Port Harcourt and Enugu. Already read by the AI, so they open instantly.

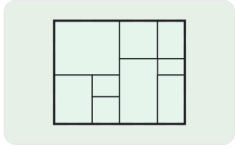
1



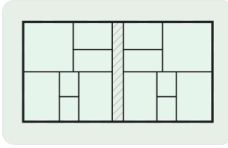
Two Bedroom Bungalow
Ikorodu, Lagos
Compact family starter home, about 96 m2, single floor.
2
[Try this plan](#)



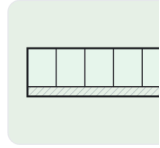
Three Bedroom Bungalow
Abuja, FCT
Three bedrooms with master en-suite and study, about 162 m2.
[Try this plan](#)



Four Bedroom Duplex
Lagos, Lagos
Two-storey family home, suspended slab, about 328 m2.
[Try this plan](#)



Block of Four 2-Bedroom Flats
Port Harcourt, Rivers
Ground plus one, two flats per floor, about 440 m2.
[Try this plan](#)



Row of Six Lock-up Shops
Enugu, Enugu
Six shops with front walk-toilets, about 144 m2.
[Try this plan](#)

- 1 Thumbnail drawn from the same room data as the PDF sheet.
- 2 Runs the sample through the normal upload pipeline.

Konstruct documentation

Page 12 of 30

6. Upload a plan

PDF or image up to 8 MB. Pressing the button without a file shows the inline error.

Konstruct My plans Upload Vendors

Upload a plan
One AI read per file. The same file is never sent twice.

1 Plan file

2 Choose a plan file

3 Project name (optional)
For example: Okafor family house, Enugu

4 Site city
Lagos

5 Get material breakdown

Vendors and prices come from this city first, then the same state and region.

No plan at hand? Try a sample

- 1 Tap to choose a PDF, PNG, JPG or WEBP of the drawing.
- 2 Inline validation message from the server action.
- 3 Optional project name. The AI suggests one if empty.
- 4 Site city. Vendors and prices come from here first.
- 5 Sends the file. New files take about 20 seconds (one AI call).

7. Plan page

Laid out like an app page in the Play Store: icon, name, stats row and one green action.

The screenshot shows the 'My plans' page in the Konstruct app. At the top, there's a navigation bar with 'Konstruct', 'My plans' (active), 'Upload', and 'Vendors'. A user profile icon 'CO' is in the top right. Below the navigation bar, a back arrow and 'My plans' link are on the left. The main content area features a plan card for 'Ogui Road shops' in Enugu, Enugu, with a reference 'KON-20261001-0002'. The card includes a stats row with '₦21.2M' (Estimated total), '37' (Materials), '144 m2' (Floor area), and a 'Needs review' status (Priced 3 days ago). Below the stats row are three buttons: 'Download PDF breakdown' (highlighted with a red box and number 3), 'View plan', and 'Call vendors'. A yellow warning box (highlighted with a red box and number 4) states: 'Check 2 lines before you buy. 2 items are not in our price list: ask a vendor and type the price.' Below this, a note says 'Prices valid until 15 Oct 2026. Confirm with the vendor before paying.' The 'Who to call' section lists suppliers near Enugu: 'Ogbete Building Materials' (Cement and Concrete, 0706 142 6754), 'Emene Blocks and Sand' (Sand, Granite and Filling, 0809 295 2529), 'Coal Camp Electrical and D...' (Doors and Windows, 0816 658 0504), and 'Ogui Roa' (Tiles and Finishes, 0812 602 2403).

- 1 Plan name, site and quote reference.
- 2 Stats row: total, number of materials, floor area, status.
- 3 Downloads the professional PDF breakdown.
- 4 Lines that need a person: items not in the price list.

8. Who to call

One or two suppliers per material category, nearest first, with call and WhatsApp buttons.

 **Konstruct**

[My plans](#) [Upload](#) [Vendors](#)

CO

Who to call

Suppliers near Enugu for the materials in this plan.

1

OM

Ogbete Building Materials

Ogbete, Enugu
3.8 ★ (379) In your city

Cement and Concrete

Delivers to site

2

0706 142 6754

3

WhatsApp

ES

Emene Blocks and Sand

Emene, Enugu
4.1 ★ (417) In your city

Sand, Granite and Filling

Delivers to site

0809 295 2529

WhatsApp

CD

Coal Camp Electrical and D...

Coal Camp, Enugu
4.3 ★ (193) In your city

Doors and Windows

0816 658 0504

OS

Ogui Roa

Ogui, Enugu
3.6 ★ (53)

Tiles and Finishes

0812 602 2405

Material breakdown

Tap a line to see how it was measured, who sells it, or to change the quantity.

Cement and Concrete	₦5,986,000
Portland cement, 50kg bag 560 bag x ₦10,300	₦5,768,000
Damp proof membrane, 1000 gauge roll 4 roll x ₦54,500	₦218,000
Sand, Granite and Filling	₦2,775,000
Sharp sand 68 tonne x ₦8,950	₦608,600
Plaster (soft) sand	₦286,900

- 1 Vendor card: initials avatar, area, rating and distance tier.
- 2 tel: link. On a phone this dials the vendor.
- 3 WhatsApp with a message that names the site city.

9. A material line, opened

Tap any line to see how it was measured, who sells it at what price, and to correct it.

Konstruct My plans Upload Vendors

1 Portland cement, 50kg bag
560 bag x ₦10,300 ₦5,768,000

2 How it was measured: Foundation footing, block infill, slab, lintels, columns, screed and mortar

3 **OM** Ogbete Building Materials
Ogbete, Enugu
3.8 ★ (379) In your city
₦10,300 per bag
Delivers to site
0706 142 6754 WhatsApp

4 Quantity (bag)
560 Save

Remove this line

Damp proof membrane, 1000 gauge roll
4 roll x ₦54,500 ₦218,000

Sand, Granite and Filling ₦2,775,000

Sharp sand
68 tonne x ₦8,950 ₦608,600

Plaster (soft) sand
38 tonne x ₦7,550 ₦286,900

- 1 Quantity x unit price = amount.
- 2 How the AI measured it, in its own short words.
- 3 Cheapest nearby vendor for this exact item and their price.
- 4 Change the quantity. A person's number replaces the AI estimate.

10. Items outside the price list

Roller shutters are not in the catalog, so the user asks a vendor and types the price.

KonstructMy plansUploadVendorsCO

Galvanised steel roller shutter 3.0 x 3.0m

6 piece

How it was measured: 6 shop front entrance roller shutter doors specified in schedule

Quantity (piece)

6

Remove this line

Unit price (Naira)

Ask a vendor

Save

Needs price

Aluminium fanlight window 3.0 x 0.4m

6 piece

Needs price

Materials subtotal

Contingency (10%)

Estimated total

Excludes 2 items without a price.

₦19,281,000

₦1,928,100

₦21,209,100

About this plan

1 Needs price: excluded from the total until priced.

2 Unit price field, only shown for items outside the catalog.

Konstruct documentation

Page 17 of 30

11. Totals and notes

Subtotal, 10% contingency and the estimated total, then the AI's summary and assumptions.

Konstruct

[My plans](#)

[Upload](#)

[Vendors](#)

CO

Remove this line

Aluminium fanlight window 3.0 x 0.4m

6 piece

Needs price

Materials subtotal

Contingency (10%)

1

Estimated total

Excludes 2 items without a price.

₦19,281,000

₦1,928,100

₦21,209,100

About this plan

Bill of quantities material take-off for a row of six lock-up shops with two rear/end water closets and a front covered walkway, based on Nigerian standard measurement practice.

2

Assumptions

- Ground condition is firm dry terrain in Enugu requiring standard 900mm foundation excavation depth.
- Shop floors receive a 1:3 cement-sand screed finish directly on oversite concrete as specified.
- No ceiling is installed in the shops per note 7; rafters and aluminium roof remain exposed.
- Walkway slab is 100mm thick concrete with a 1:3 cement screed finish.
- Prepaid meters and specialized connection hardware are excluded as contractor/utility items.

3

Plan file

Read by

Added

Labour

shops-6-ogui-enugu.pdf

google/gemini-3.8-flash

1 Oct 2026

Not included

Delete plan

- 1 Estimated total in Naira.
- 2 Assumptions the AI made, so a builder can check them.
- 3 Plan file, model used and the date added.

12. Stale prices

Plans priced more than 30 days ago warn the user. One button re-prices from today's vendor lists.

The screenshot shows the Konstruct web application interface. At the top, there's a navigation bar with 'Konstruct' logo, 'My plans', 'Upload', and 'Vendors' links. A user profile icon 'CO' is in the top right. Below the navigation bar, the 'My plans' section is active, showing a plan titled 'Ogunleye starter home' located in 'Ikorodu, Lagos'. The plan details include 'Bungalow with 2 bedrooms, 1 floor. Ref KON-20260819-0005'. Key metrics are displayed: '₦16.6M Estimated total', '41 Materials', and '96 m2 Floor area'. A 'Prices stale' badge indicates the plan was 'Priced 46 days ago'. Below these metrics are three buttons: 'Download PDF breakdown', 'View plan', and 'Call vendors'. A red warning banner (1) states 'Prices are out of date. Priced on 19 Aug 2026. Market prices move fast; refresh to use today's vendor prices.' and includes a 'Refresh prices' button (2). Below the banner, the 'Who to call' section lists suppliers near Ikorodu: 'Ikorodu Builders Mart' (4.7 stars, 375 reviews, in city), 'Coker Builders Depot' (4.2 stars, 355 reviews, in state), 'Agric Blocks Industry' (4.9 stars, 481 reviews, in city), and 'Ebute Ro' (4.6 stars, 273 reviews).

- 1 Banner shows the pricing date.
- 2 Refresh prices: no AI call, only the pricing step runs again.

13. Vendor directory

All listed suppliers, filtered by city and by material category.

The screenshot shows the 'Vendors' page in the Konstruct application. At the top, there's a navigation bar with 'Konstruct', 'My plans', 'Upload', and 'Vendors' (highlighted). A user profile icon 'CO' is in the top right. Below the navigation, the title 'Vendors' is followed by the text '34 building material suppliers in 8 cities.'.

Two filter rows are present:

- City filters (1):** A row of chips including 'All cities', 'Lagos' (selected), 'Ikorodu', 'Ibadan', 'Abuja', 'Port Harcourt', 'Benin City', 'Enugu', and 'Kano'.
- Material category filters (2):** A row of chips including 'All materials' (selected), 'Cement and Concrete', 'Sand, Granite and Filling', 'Blocks', 'Iron Rods and Steel', 'Timber', 'Roofing', and 'Doors and Windows'.

Below the filters, a grid of vendor cards is displayed. Each card includes:

- Brand logo and name:** A colored square with initials followed by the vendor's name.
- Location:** The city and state (e.g., 'Surulere, Lagos').
- Rating and reviews:** A star rating and the number of reviews (e.g., '4.8 ★ (415)').
- Material categories:** A green chip labeled 'In your city'.
- Services:** A list of materials or services provided.
- Delivery status:** A truck icon and the text 'Delivers to site'.
- Contact buttons:** A green button with a phone icon and a WhatsApp button.

Red numbered callouts point to these elements:

- 1:** Points to the 'Lagos' city chip.
- 2:** Points to the 'All materials' category chip.
- 3:** Points to the 'Surulere Paints Centre' vendor card.

- 1 City chips. Defaults to the user's city.
- 2 Material category chips.
- 3 Brands stocked, rating, delivery and the call buttons.

14. Account

Shows the user's generated avatar, their default city and sign out.

Konstruct My plans Upload Vendors

1 **CO** Chidinma Okafor
demo@konstruct.ng

2 Default city for new plans
Lagos

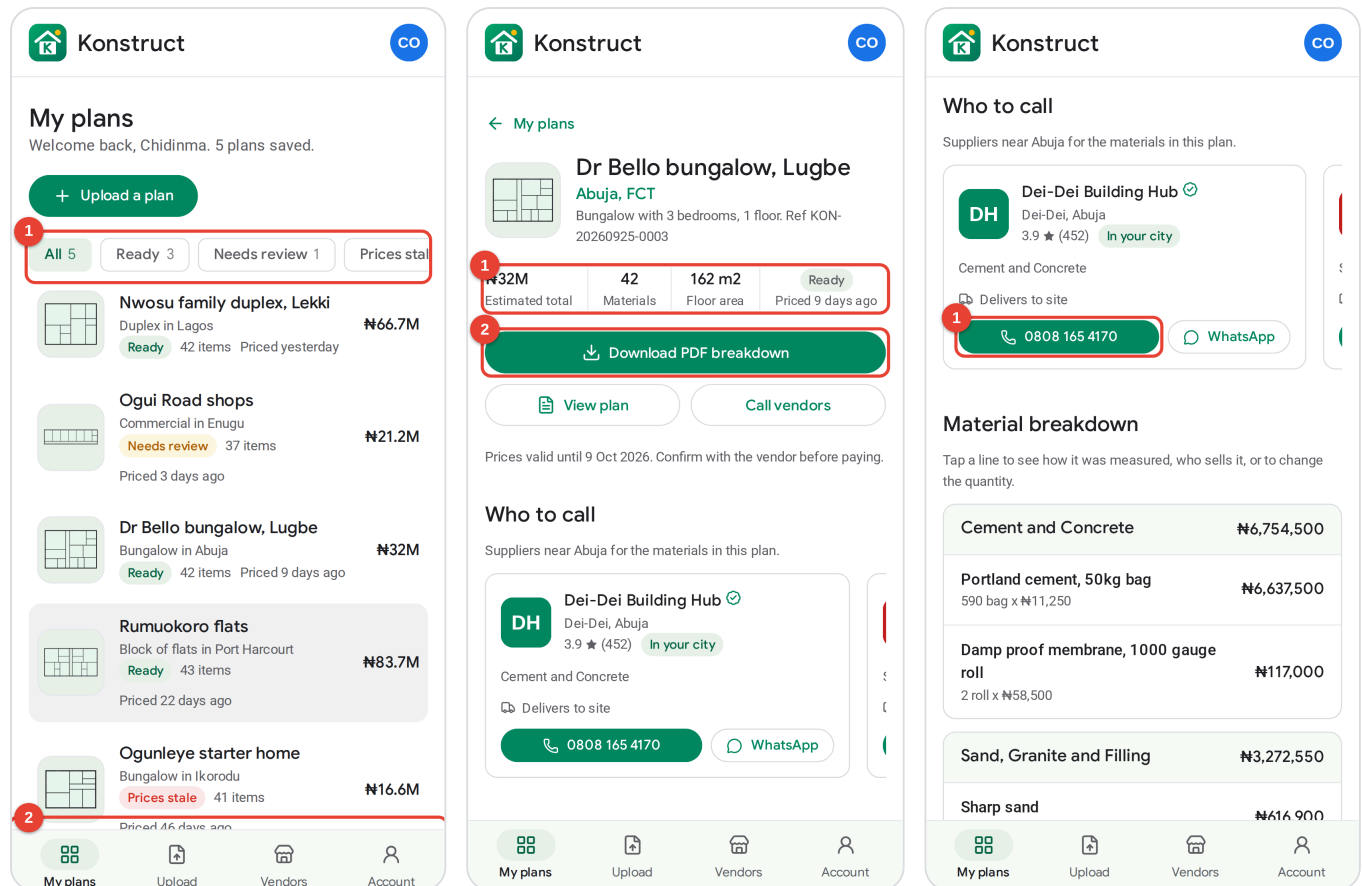
Save city

3 [Sign out](#)

- 1 Initials avatar made locally with DiceBear.
- 2 Default city for new uploads and the vendor list.
- 3 Sign out clears the session cookie.

5. Mobile view

On screens under 720px the top tabs become a bottom navigation bar, carousels become swipeable, the primary button goes full width and all grids use `minmax(0, 1fr)` so long names never push the page sideways. Shown on a Pixel 7 profile (412 x 915).



Phone: My plans

- 1 Chips scroll sideways.
- 2 Bottom navigation, like the Play Store app.

Phone: plan page

- 1 Stats row scrolls if it is wider than the phone.
- 2 Full-width primary action.

Phone: call a vendor

- 1 Tap to dial.

6. The PDF breakdown

The Download PDF button streams an A4 estimate built on the server. Page 1 has the quote reference, validity date, client and plan details, the totals in four boxes and the start of the bill of materials grouped by category. The vendor pages list up to three suppliers per category with address, distance tier, rating and phone number. The last part holds the AI's summary, its assumptions and a plain disclaimer.

Konstrukt

MATERIAL ESTIMATE

Ref KON-20261003-0001

Priced 3 Oct 2026, valid until 17 Oct 2026

Nwosu family duplex, Lekki

Duplex, 4 bedrooms, 2 floors, about 328 m2. Site: Lagos, Lagos.

PREPARED FOR

Chidinma Okafor

EMAIL

demo@konstruct.ng

PLAN FILE

duplex-4bed-lekki.pdf

STATUS

Ready, issued 4 Oct 2026

ESTIMATED TOTAL

N66,659,560

MATERIALS

N60,599,600

CONTINGENCY 10%

N6,059,960

LINE ITEMS

42

Bill of materials

#	Description	Qty	Unit	Rate (NGN)	Amount (NGN)
Cement and Concrete					N13,208,000
1	Portland cement, 50kg bag	1,150	bag	10,500	12,075,000
2	Plaster of Paris (POP), 40kg bag	140	bag	6,550	917,000
3	Damp proof membrane, 1000 gauge roll	4	roll	54,000	216,000
Sand, Granite and Filling					N6,115,250
4	Sharp sand	125	tonne	8,750	1,093,750
5	Plaster (soft) sand	65	tonne	7,600	494,000
6	Granite chippings, 3/4 inch	185	tonne	20,500	3,792,500
7	Laterite filling	175	tonne	4,200	735,000
Blocks					N4,902,500
8	Sandcrete block, 9 inch (225mm) hollow	3,850	piece	840	3,234,000
9	Sandcrete block, 6 inch (150mm) hollow	2,350	piece	710	1,668,500
Iron Rods and Steel					N10,960,600
10	Iron rod, 10mm x 12m length	185	length	9,400	1,739,000
11	Iron rod, 12mm x 12m length	360	length	14,500	5,220,000
12	Iron rod, 16mm x 12m length	145	length	25,000	3,625,000
13	Binding wire, 20kg roll	6	roll	38,500	231,000
14	Nails, assorted sizes	80	kg	1,820	145,600
Timber					N4,884,500
15	Hardwood 2x4 inch, 12ft	420	piece	5,000	2,100,000

Konstrukt

MATERIAL ESTIMATE

Ref KON-20261003-0001

Priced 3 Oct 2026, valid until 17 Oct 2026

16	Hardwood 2x6 inch, 12ft	190	piece	7,750	1,472,500
17	Formwork plank 1x12 inch, 12ft	320	piece	4,100	1,312,000
Roofing					N3,721,000
18	Stone-coated roofing tile sheet	520	sheet	7,050	3,666,000
19	Roofing nails	25	kg	2,200	55,000
Doors and Windows					N3,830,000
20	Steel security door, external	2	piece	240,000	480,000
21	Hardwood panel door with frame, internal	14	piece	82,500	1,155,000
22	Aluminium sliding window, 1.2 x 1.2m	22	piece	92,000	2,024,000
23	Aluminium louvre window, 0.6 x 0.6m	6	piece	28,500	171,000
Tiles and Finishes					N6,572,750
24	Porcelain floor tiles, 60 x 60cm	345	m2	11,650	4,019,250
25	Ceramic wall tiles, 30 x 60cm	165	m2	10,000	1,650,000
26	Tile adhesive, 20kg bag	130	bag	6,950	903,500
Paint					N2,082,500
27	Emulsion paint, 20L bucket	28	bucket	42,500	1,190,000
28	Textured exterior paint, 20L bucket	15	bucket	59,500	892,500
Plumbing					N1,838,300
29	PVC soil pipe, 4 inch x 5.8m	24	length	11,200	268,800
30	PPR water pipe, 20mm x 4m	40	length	3,650	146,000
31	Water closet (WC) set, complete	5	piece	101,000	505,000
32	Wash hand basin with pedestal	5	piece	47,500	237,500
33	Kitchen sink, stainless steel	1	piece	57,000	57,000
34	Shower mixer set	4	piece	36,500	146,000
35	Plastic water tank, 2000 litres	2	piece	239,000	478,000
Electrical					N2,484,200
36	Single core cable 1.5mm, 100m roll	10	roll	54,500	545,000
37	Single core cable 2.5mm, 100m roll	14	roll	83,500	1,169,000
38	PVC conduit pipe, 20mm x 3m	120	length	600	72,000
39	13A double switched socket	40	piece	3,550	142,000
40	1-gang light switch	30	piece	2,500	75,000
41	LED ceiling light fitting	48	piece	8,150	391,200

Konstrukt material estimate, ref KON-20261003-0001

Page 1 of 6

Konstrukt material estimate, ref KON-20261003-0001

Page 2 of 6

42	Distribution board, 8-way	2 piece	45,000	90,000
			Materials subtotal	₦60,599,600
			Contingency (10%)	₦6,059,960
			Estimated total	₦66,659,560

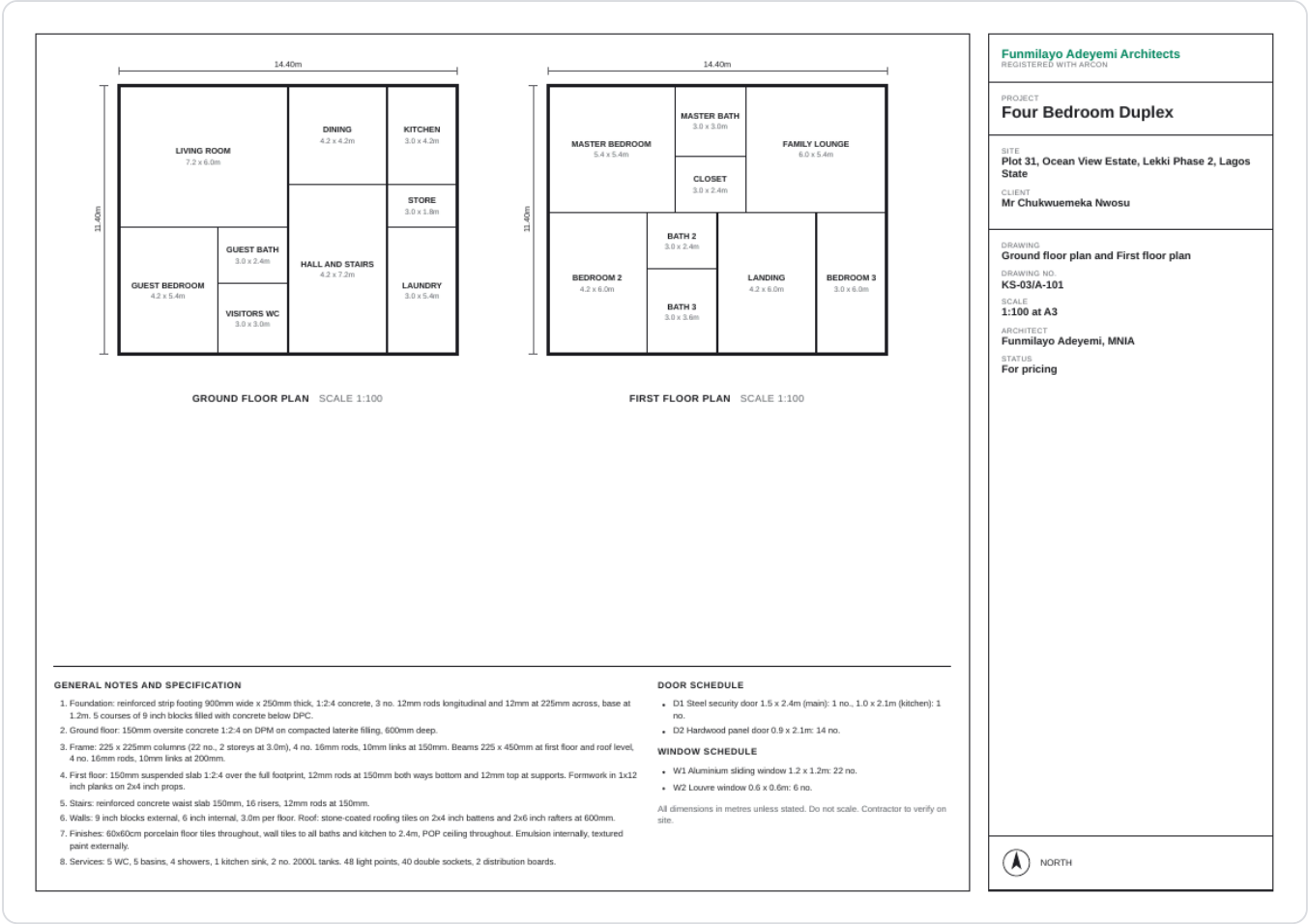
Vendors to call

Nearest suppliers to Lagos for each category, best rated first. Quote reference KON-20261003-0001 when you call.

Cement and Concrete				
Coker Builders Depot (verified)	In your city	4.2 / 5	0706 696 7884	
Shop 12, Coker Building Materials Market, Orile, Lagos				
Ikorodu Builders Mart (verified)	In your state	4.7 / 5	0802 231 1383	
112 Sagamu Road, Ikorodu				
Iwo Road Cement Depot (verified)	In your region	4.6 / 5	0818 975 3391	
Iwo Road Interchange, Ibadan				
Sand, Granite and Filling				
Badore Sand and Granite (verified)	In your city	4.0 / 5	0701 710 8326	
14 Badore Road, Ajah, Lagos				
Agric Blocks Industry (verified)	In your state	4.9 / 5	0816 827 3322	
Agric Bus Stop, Ikorodu				
Ikorodu Builders Mart (verified)	In your state	4.7 / 5	0802 231 1383	
112 Sagamu Road, Ikorodu				
Blocks				
Badore Sand and Granite (verified)	In your city	4.0 / 5	0701 710 8326	
14 Badore Road, Ajah, Lagos				
Agric Blocks Industry (verified)	In your state	4.9 / 5	0816 827 3322	
Agric Bus Stop, Ikorodu				
Ikorodu Builders Mart (verified)	In your state	4.7 / 5	0802 231 1383	
112 Sagamu Road, Ikorodu				
Iron Rods and Steel				
Coker Builders Depot (verified)	In your city	4.2 / 5	0706 696 7884	
Shop 12, Coker Building Materials Market, Orile, Lagos				
Ikorodu Builders Mart (verified)	In your state	4.7 / 5	0802 231 1383	
112 Sagamu Road, Ikorodu				
Ebute Roofing and Steel	In your state	4.6 / 5	0818 599 3139	
5 Ebute Road, Ikorodu				
Timber				
Oko Baba Timber Traders (verified)	In your city	4.0 / 5	0809 899 3487	
Oko Baba Sawmill, Ebute Metta, Lagos				
Ebute Roofing and Steel	In your state	4.6 / 5	0818 599 3139	
5 Ebute Road, Ikorodu				
Bodija Timber and Roofing (verified)	In your region	3.8 / 5	0815 647 7289	
Bodija Market Road, Ibadan				

7. Sample plans and AI results

Five A3 drawing sheets live in public/samples/. They are generated by `npm run samples:make` from room data, so the floor plans, notes, door and window schedules all agree. Each one was sent once through the real AI call (`npm run samples:extract`) and the answers are stored in `src/lib/sample-extractions.json` and seeded into the cache.



Sample	Site	Area m2	Lines	Cement bags	Blocks	Rod lengths
Two Bedroom Bungalow	Ikorodu, Lagos	96	41	380	2,270	78
Three Bedroom Bungalow	Abuja, FCT	162	42	590	4,600	136
Four Bedroom Duplex	Lagos, Lagos	328	42	1,150	6,200	690
Block of Four 2-Bedroom Flats	Port Harcourt, Rivers	441	43	1,450	7,700	900
Row of Six Lock-up Shops	Enugu, Enugu	144	37	560	2,950	120

The unit tests check these answers: floor area within 20% of the drawn footprint, 2 to 7 bags of cement per m2, bigger buildings need more cement, window and door counts match the schedules, only the suspended-slab buildings get 16mm rods, and the shops keep roller shutters as an item to price by hand.

8. Running locally

```
# Postgres (re-check it is up before tests; containers can restart it)
service postgresql start && pg_isready
createdb konstrukt && createdb konstrukt_test && createdb konstrukt_e2e

cp .env.example .env          # set DATABASE_URL, SESSION_SECRET, OPENROUTER_API_KEY
npm install
npm run db:migrate
npm run db:seed               # vendors, demo user and plans dated from today
npm run dev                   # http://localhost:3000, demo@konstrukt.ng / demo1234
```

Script	What it does
npm run build / start	Production build and server
npm run db:generate	New SQL migration from schema changes
npm run db:reset	Wipe and seed again
npm run samples:make	Re-render the five sample PDFs and thumbnails
npm run samples:extract	Send the samples through the real AI once
npm run docs:pdf	Rebuild this document (after npm run build)

Environment variables

Name	Use
DATABASE_URL	Postgres connection
SESSION_SECRET	Signs session tokens (required in production)
OPENROUTER_API_KEY, OPENROUTER_MODEL, OPENROUTER_REASONING	AI access, model choice and reasoning effort (default low)
KONSTRUCT_TODAY	Pins today's date (YYYY-MM-DD) for demos and tests
KONSTRUCT_AI_MODE=mock	Deterministic fake AI for tests
TEST_DATABASE_URL, E2E_DATABASE_URL	Separate databases for integration and e2e tests

9. Testing

Suite	Tool	Tests	What it covers
Unit	Vitest	52	Dates, money, catalog, normalizing AI answers, vendor ranking, pricing, statuses, validation, JWT, and the five real sample take-offs
Integration	Vitest + real Postgres	15	Seed mix, user scoping, cache hit without network, mock AI call then cache, non-plan rejection, edits, refresh prices, delete, PDF rendering, database ping
End to end (desktop)	Playwright, Chromium	13	Sign in, sign up errors, filters, plan page, PDF download, refresh prices, pricing by hand, upload errors and success, sample carousel, vendor filters, another user's plan returns 404
End to end (mobile)	Playwright, Pixel 7	1	Bottom nav, no sideways scroll, tap-to-call link
Total		81	All passing

```
npm test           # unit + integration (needs TEST_DATABASE_URL)
npm run test:e2e   # builds must exist: npm run build first
```

E2E tests run the production build against `konstruct_e2e` with `KONSTRUCT_TODAY=2026-10-04` and `KONSTRUCT_AI_MODE=mock`, so no tokens are spent and the dates never drift.

10. Deployment

Railway project "school-projects" holds two services for this app: **konstruct** (Node, built from the GitHub repository) and **Postgres-moRa** (PostgreSQL).

Setting	Value
Source	GitHub emmanuelekopimo/konstruct, auto-deploy on push
Build	npm run build (Railpack, Node 22)
Start	tsx scripts/migrate.ts && tsx scripts/seed.ts --if-empty && next start -H 0.0.0.0 -p \$PORT
Health check	/api/health runs select 1 on the database and returns {"status":"ok","database":"ok"}
Variables	DATABASE_URL=\${Postgres-moRa.DATABASE_URL}, SESSION_SECRET (random 64 hex), NODE_ENV=production, OPENROUTER_API_KEY, OPENROUTER_MODEL
Public URL	https://konstruct-production-c62c.up.railway.app

The seed only runs when the users table is empty, so redeploys never overwrite real data. All config is in `railway.json` in the repository.

11. 5-minute presentation script

Time	Say	Do
0:00 - 0:40	"Anyone who has built a house in Nigeria knows the problem: you have a drawing, but you do not know how many bags of cement or blocks to buy, or who sells them near your site. Konstruct turns the drawing into a priced list and gives you vendors to call."	Open the landing page. Point at the hero and the sample sheets.
0:40 - 1:20	"I sign in with the demo account. These are Chidinma's plans. Green means ready, amber means a person must check something, red means the prices are old."	Press Sign in (pre-filled). Tap the Needs review and Prices stale chips.
1:20 - 2:20	"Let me upload a new plan. This is an architect's sheet for a three-bedroom bungalow in Lugbe, Abuja. The AI reads it once and returns only quantities; everything about money is our own code. Because this file was read before, the answer comes from the cache and costs nothing."	Upload, choose public/samples/bungalow-3bed-lugbe.pdf (or press Try this plan), city Abuja, submit.
2:20 - 3:20	"Here is the result: about 590 bags of cement, 4,600 blocks, rods, roofing, doors, tiles and fittings, priced at about 32 million Naira. Open cement: this is how it was measured, and here is the cheapest vendor in Abuja. One tap calls them, or sends a WhatsApp message with the quantity and site already written."	Scroll the breakdown. Open the cement line. Show Call and WhatsApp.
3:20 - 4:00	"The client can download a professional PDF to send to a builder or a bank."	Press Download PDF. Show page 1 and the vendor page.
4:00 - 4:30	"Two rules worth showing: old prices go stale after 30 days and refresh without another AI call; and items not in our list, like these roller shutters, wait for the user to type a price."	Open Ogunleye starter home, press Refresh prices. Open Ogui Road shops, show Needs price.
4:30 - 5:00	"It works on a phone, it is tested with 81 automated tests, and it is live on Railway. Questions?"	Show the phone view (browser dev tools) or the mobile screenshots.

Backup plan: if the network is slow, every step above uses cached sample plans, so nothing waits on the AI.

12. Decisions and limits

- **Vendor listings are demo data.** The markets, areas and brands are real, but business names and phone numbers are illustrative and must be replaced with verified vendors before real use.
- **Prices** are typical 2026 Nigerian market prices for each item, adjusted per city and per vendor. They are estimates, not quotes.
- **Materials only.** Labour, transport and VAT are not included; a 10% contingency is added.
- **The AI only measures.** It never prices or picks vendors, which keeps it to one short call per file and makes the money logic testable.
- **Cache by file hash.** Uploading the same bytes again gives the same answer for free.
- **Files are stored in Postgres** (bytea) because Railway disks are not persistent. Fine for an 8 MB limit and a demo.
- **Quantities are a first pass** for planning and budgeting. A quantity surveyor should confirm them before buying.